

Solution 4: VaR and ES Calculations

ME317: Statistical Methods for Risk Management

Question 1 — Expected Shortfall of an insurance loss

Loss L with $\mathbb{P}(L \leq l) = 1 - l^{-2}$ for $l \geq 1$. Compute $\text{ES}_{0.99}(L)$.

Using $\text{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u(L) du$ with $\alpha = 0.99$, first find the quantile. Let $q_u := \text{VaR}_u(L)$; then $\mathbb{P}(L \leq q_u) = u$ gives

$$1 - \frac{1}{q_u^2} = u \implies q_u = (1 - u)^{-1/2}.$$

Hence

$$\text{ES}_{0.99}(L) = 100 \int_{0.99}^1 (1 - u)^{-1/2} du = 100 \left[-2(1 - u)^{1/2} \right]_{0.99}^1 = 200(0.01)^{1/2} = 200 \times 0.1 = 20.$$

Question 2 — Linearised VaR of three strategies

Daily log-returns $(X_1, X_2, X_3)' \sim N_3(0, \Sigma)$, prices $S_1 = S_2 = S_3 = 100$, returns independent across days, and

$$\Sigma = \begin{pmatrix} 10^{-4} & 0.5 \times 10^{-4} & 0 \\ 0.5 \times 10^{-4} & 10^{-4} & -0.5 \times 10^{-4} \\ 0 & -0.5 \times 10^{-4} & 10^{-4} \end{pmatrix}.$$

Total capital 300, fully invested with weights w_i . Writing λ_i for the number of shares in stock i , $\lambda_i S_i = 300 w_i$, so $\lambda_i = 3w_i$. The one-day loss is

$$L_{t+1} = - \sum_i \lambda_i S_{t,i} (e^{X_{t+1,i}} - 1), \quad L^\Delta = - \sum_i \lambda_i S_{t,i} X_{t+1,i} = -300 \sum_i w_i X_{t+1,i},$$

using the linearisation $e^x - 1 \approx x$.

(a) Linearised losses. Dropping the day index (one-day case), with X_i the return of stock i :

$$L_A^\Delta = -150(X_1 + X_2), \quad L_B^\Delta = -150(X_2 + X_3), \quad L_C^\Delta = -100(X_1 + X_2 + X_3).$$

(For A and B, $\lambda_i S_i = 300 \cdot \frac{1}{2} = 150$; for C, $300 \cdot \frac{1}{3} = 100$.)

(b) VaR ratios. The relevant variances (recall $\text{Var}(U + V) = \sigma_U^2 + \sigma_V^2 + 2\text{Cov}(U, V)$):

$$\begin{aligned} \text{Var}(X_1 + X_2) &= 10^{-4} + 10^{-4} + 2(0.5 \times 10^{-4}) = 3 \times 10^{-4}, \\ \text{Var}(X_2 + X_3) &= 10^{-4} + 10^{-4} + 2(-0.5 \times 10^{-4}) = 10^{-4}, \\ \text{Var}(X_1 + X_2 + X_3) &= 3 \times 10^{-4} + 2(0.5 \times 10^{-4} + 0 - 0.5 \times 10^{-4}) = 3 \times 10^{-4}, \end{aligned}$$

where the last uses $\text{Cov}(X_1, X_3) = 0$. Since each loss is a scaled centred normal, $\text{VaR}_\alpha = (\text{scale}) \times \text{VaR}_\alpha(Z)$ with $Z \sim N(0, 1)$:

$$\text{VaR}_\alpha(L_A^\Delta) = 1.5\sqrt{3} \text{VaR}_\alpha(Z), \quad \text{VaR}_\alpha(L_B^\Delta) = 1.5 \text{VaR}_\alpha(Z), \quad \text{VaR}_\alpha(L_C^\Delta) = \sqrt{3} \text{VaR}_\alpha(Z).$$

Therefore

$$\frac{\text{VaR}_{0.99}(L_B^\Delta)}{\text{VaR}_{0.99}(L_A^\Delta)} = \frac{1.5}{1.5\sqrt{3}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}, \quad \frac{\text{VaR}_{0.99}(L_C^\Delta)}{\text{VaR}_{0.99}(L_A^\Delta)} = \frac{\sqrt{3}}{1.5\sqrt{3}} = \frac{2}{3}.$$

(c) Which is riskiest, and why. Strategy **A** is the most risky (largest VaR): it combines stocks 1 and 2, which are *positively* correlated, so their moves reinforce each other. Strategy **B** uses stocks 2 and 3, which are *negatively* correlated — a natural hedge that cancels much of the variation, giving the lowest VaR. Strategy **C** diversifies across all three, reducing risk relative to A but not as effectively as the hedge in B.

(d) Meeting a regulatory 10-day VaR limit. The limit is 20% of $V_0 = 300$, i.e. 60. Take the least-risky portfolio B. Its 10-day linearised loss (10 independent days) is

$$L_{B,10}^\Delta = -150 \left(\sum_{i=1}^{10} X_{t+i,2} + \sum_{i=1}^{10} X_{t+i,3} \right), \quad \mathbb{E}[L_{B,10}^\Delta] = 0,$$

with variance

$$\begin{aligned} \text{Var}(L_{B,10}^\Delta) &= 150^2 \left(10 \cdot 10^{-4} + 10 \cdot 10^{-4} + 2 \sum_{i=1}^{10} \text{Cov}(X_{t+i,2}, X_{t+i,3}) \right) \\ &= 150^2 (10^{-3} + 10^{-3} + 2 \cdot 10 \cdot (-0.5 \times 10^{-4})) = 150^2 \cdot 10^{-3} = 1.5^2 \cdot 10 = 22.5. \end{aligned}$$

So $L_{B,10}^\Delta \sim N(0, 1.5^2 \cdot 10)$ and

$$\text{VaR}_{0.99}(L_{B,10}^\Delta) = 1.5\sqrt{10} \Phi^{-1}(0.99) \approx 1.5\sqrt{10} \times 2.3 \approx 10.9 < 60,$$

so portfolio B satisfies the constraint.